PS-0 run report: baseline, host manifest and verdict corpus

PDP oracle speedup program, `research/pdp_oracle_speedup_20261011/` — run of 2026-10-11, 21:07 UTC

1. What ran

`baseline_v1/run_ps0.sh` on `main` @ `4b6e09b78` plus this PR's diff (Conductor task T-4). Reference: `semaev_decomp::decompose` (one-word field, `gcd(q, L_V mod q)`), harness `examples/pdp_speedup`. Rungs $l \in \{8, \ldots, 12\}$, $n = 3l$, curve $y^2 + xy = x^3 + x^2 + 1$ (slugs in the baseline README). Host: Intel Xeon @ 2.80 GHz KVM guest, 4 vCPU, AVX-512F/BW, PCLMULQDQ, no cpufreq control. Isolation: `tools/isolated_bench.py run --cpus 3` (a shared container, not the repository's receipt service), 0 of 50 timed runs marked contended; one refused start (pre-flight, my own polling) with no run and no record. Corpus: 64 uniform and 16 planted targets per rung, SplitMix64 seed 20261011 + l, regenerated byte-identical in a second run. Raw records: `baseline_v1/raw/{times,iso}.jsonl`, `baseline_v1/opcount/counts.jsonl`, `baseline_v1/callgrind/`.

2. Observations

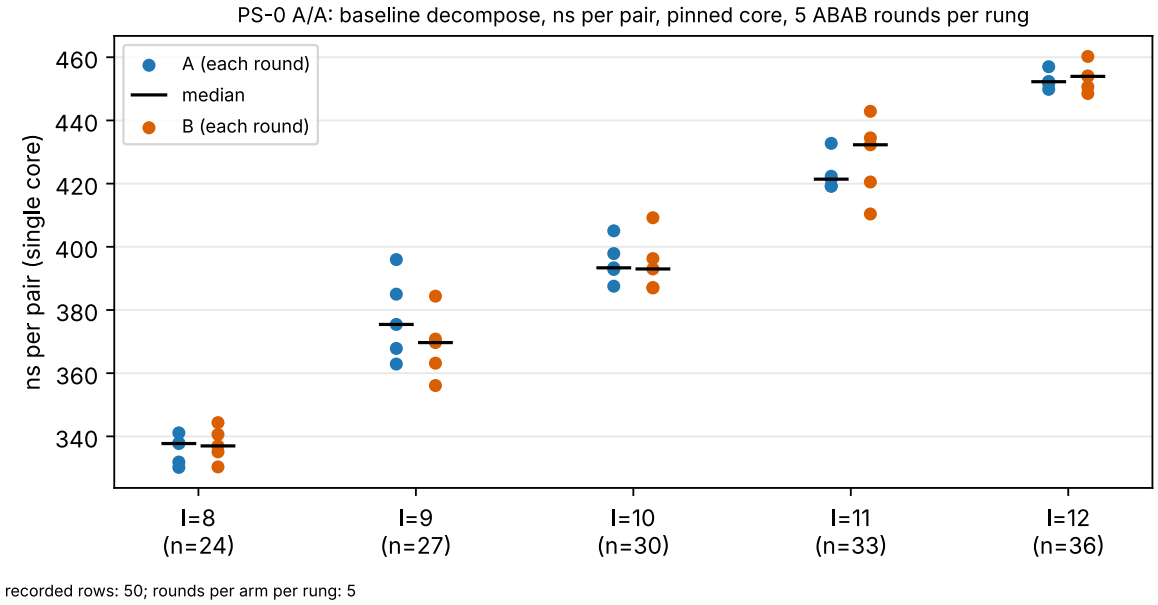


Figure 1: Wall nanoseconds per pair of the baseline, arm A and arm B being the same build under two names, alternating ABAB, five rounds per rung, pinned core 3 (50 runs; source `baseline_v1/raw/times.jsonl`). Black bars are per-arm medians.

l	A median ns/pair	B median ns/pair	A spread	B/A median	B/A range
8	337.7	337.0	3.24%	1.0085	0.9918–1.0206
9	375.4	369.7	8.81%	0.9847	0.9364–1.0008
10	393.3	393.0	4.46%	0.9986	0.9783–1.0285
11	421.4	432.3	3.22%	1.0236	0.9717–1.0564
12	452.3	453.9	1.58%	0.9979	0.9917–1.0176

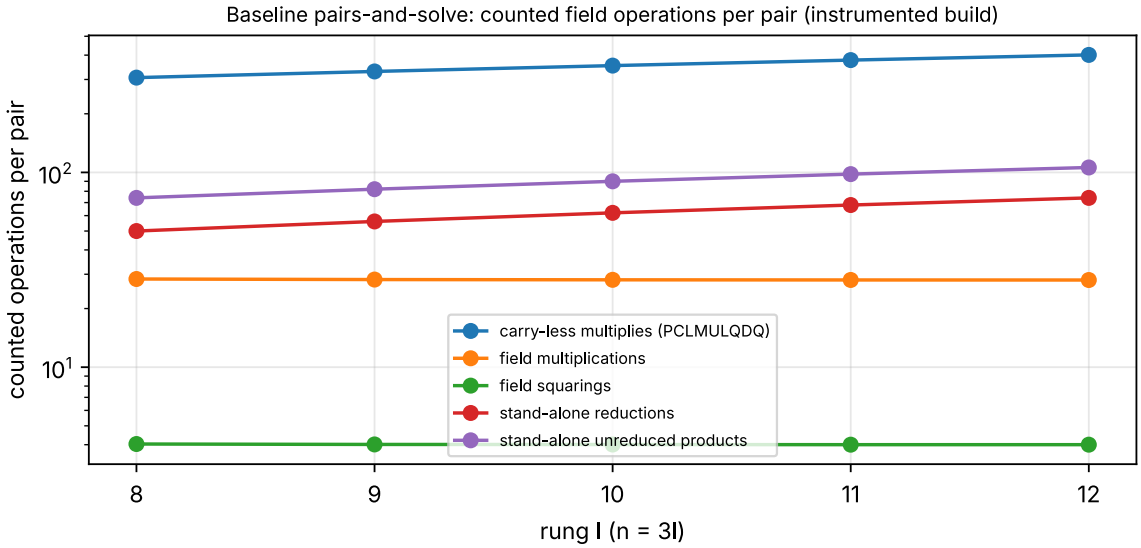


Figure 2: Counted operations per pair of the baseline (instrumented build `pdp-opcount`, 2 refuted targets per rung; source `baseline_v1/opcount/counts.jsonl`). Log scale.

l	pairs	retired instructions per pair (callgrind)	PCLMULQDQ per pair (derived from counts)
8	32,896	2356	306.6
9	131,328	2570	329.8
10	524,800	not run	353.4
11	2,098,176	not run	377.2
12	8,390,656	not run	401.1

Verdict and root-set digests are identical in three replays per rung (`baseline_v1/digests.txt`); root sets agree with a brute-force walk of every sorted triple on all 80 targets at $l = 8$ and the first 16 at $l = 9$. All 153 roots found over the five rungs re-add to $\pm R$ in the group.

3. Missing or not done

Brute-force cross-check at $l \geq 10$ not run. Callgrind at $l \geq 10$ not run (only $l = 8, 9$ were requested). The baseline binary is hashed in `manifest.json` but not committed. No speedup is claimed; this report records the reference only.